SUNWAY UNIVERSITY • SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

Autonomous Wildfire Risk Assessment & UAV Surveillance Dispatch

Mamdani Fuzzy Logic Inference Interfaced with NVIDIA Isaac Sim Digital Twin

CSC3034 Computational Intelligence • Project Supervisor: **Dr. Tang Tiong Yew**

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BACKGROUND & MOTIVATION

The Wildfire Challenge & Sensor Noise

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Limitations of Conventional Systems

- **Climate Amplification:** Megafires are rising in intensity and speed, demanding autonomous aerial detection (UN SDGs 13 & 15).
- **Binary Alarm Failures:** Simple thresholds (e.g. `T > 40°C`) cause false alarms during humid heat and miss hazards during dry, windy conditions.
- **Non-Linear Dynamics:** Fire spread is governed by interconnected atmospheric variables: heat, desiccated fuel, and turbulent wind shear.
- **Turbulence & Sensor Noise:** Low-cost drone sensors exhibit drift and electromagnetic jitter in harsh forest terrain.

Proposed Soft-Computing Solution

- **Mamdani Fuzzy Logic:** Implements continuous, robust reasoning under sensor uncertainty with human-interpretable linguistic rules.
- **Multi-Sensor Fusion:** Fuses Ambient Temperature, Relative Humidity, and Wind Speed into a normalized **Fire Risk Index (FRI)**.
- **Closed-Loop Drone Modulation:** Automatically throttles UAV flight speed (6.5 to 22.5 m/s) to maximize surveillance dwell time over hotspots.
- **NVIDIA Isaac Sim Twin:** Physics-accurate OpenUSD simulation enables safe, zero-risk validation before real-world flight.

METHODOLOGY & WORKFLOW

End-to-End Autonomous Surveillance Pipeline

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1. Sensing

UAV synthetic thermal and atmospheric probes ingest *T*, *RH*, and *WS* at 60 Hz from Isaac Sim.

2. Fuzzification

Converts crisp telemetry into membership degrees across 4 linguistic sets per input.

3. Inference

27 Mamdani min-max rules compute aggregated fuzzy consequence surfaces.

4. Defuzzification

Centroid calculation produces crisp Fire Risk Index (0-100%) and Patrol Speed (5-25 m/s).

5. Actuation

USD attributes synchronize UAV flight keyframes and dispatch instant ground alerts.

FUZZY ENGINE SPECIFICATION

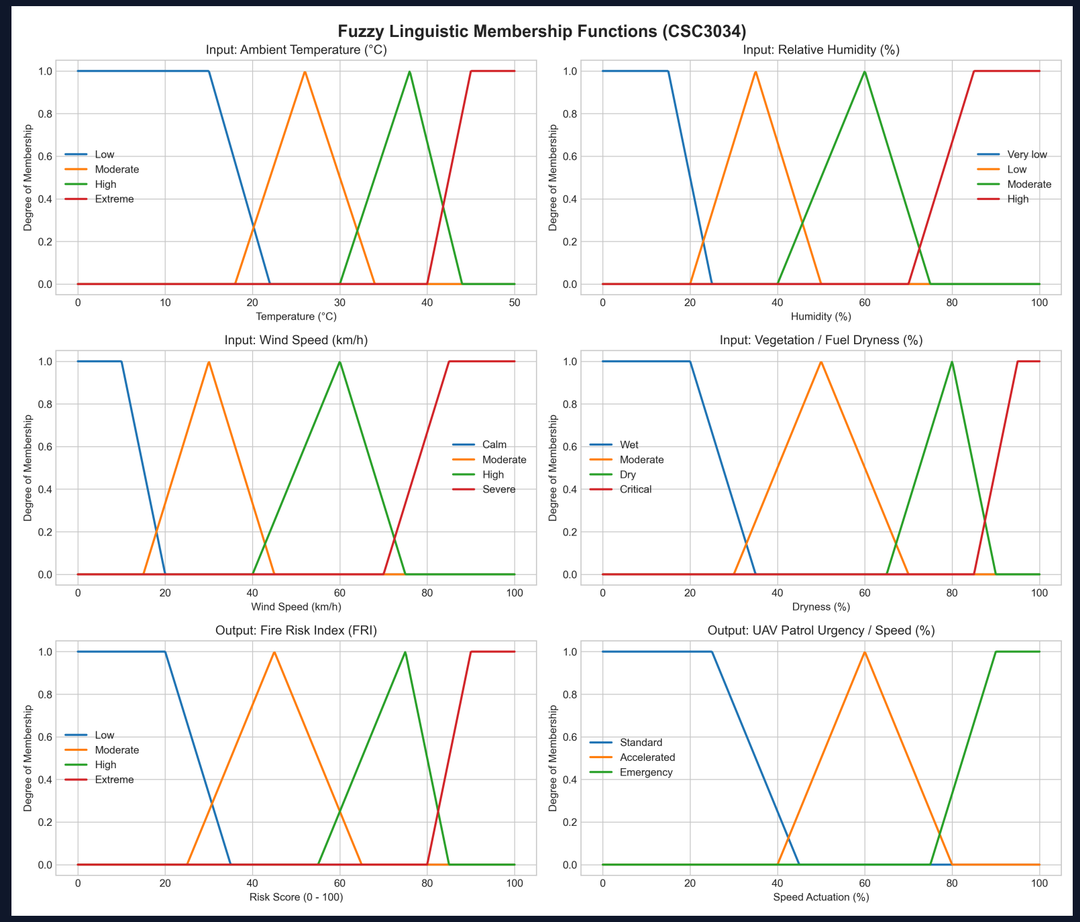
Linguistic Variables & Membership Functions

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Universes of Discourse

- **Temperature (T):** [0, 50]°C • *Low, Moderate, High, Extreme*
- **Relative Humidity (RH):** [0, 100]% • *High, Moderate, Low, Very Low*
- **Wind Speed (WS):** [0, 80] km/h • *Calm, Moderate, High, Severe*
- **Fire Risk Index (FRI):** [0, 100]% • *Safe, Caution, High, Critical*
- **Patrol Urgency Speed:** [5, 25] m/s • *Loiter, Nominal, Fast, Sprint*

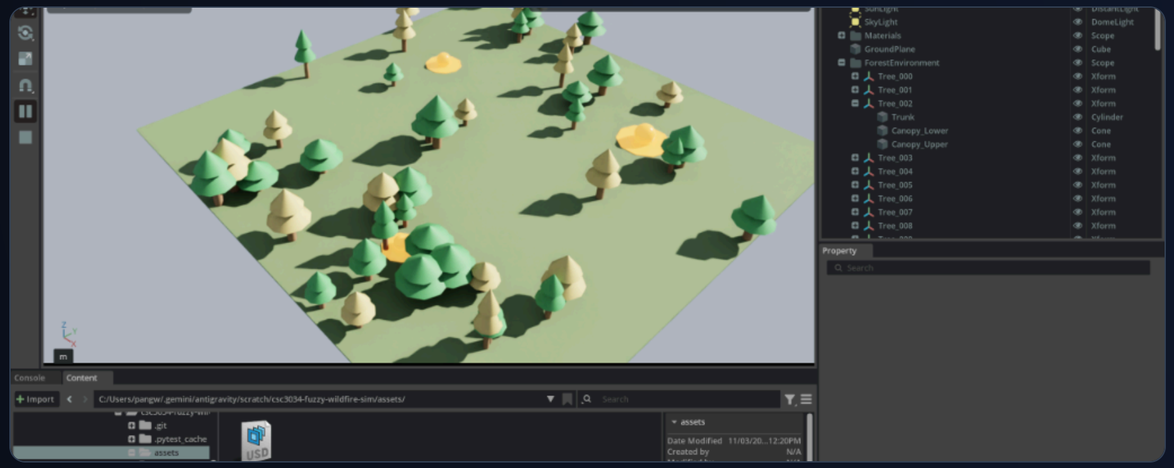
Design Principle: Trapezoidal functions for boundary domains (e.g. Extreme heat > 45°C) guarantee full saturation, while triangular functions handle transitional regions.

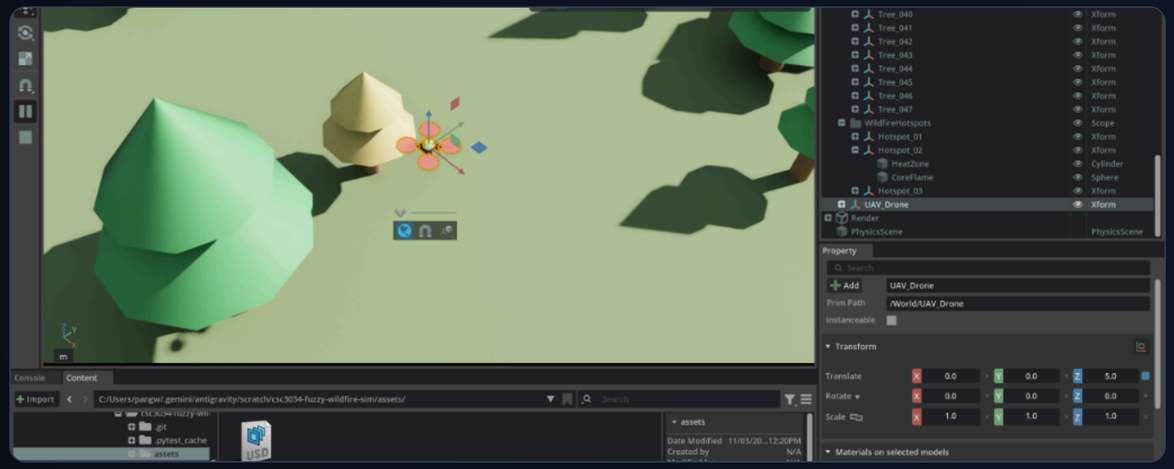


3D DIGITAL TWIN

NVIDIA Isaac Sim 6.2.0 OpenUSD Environment

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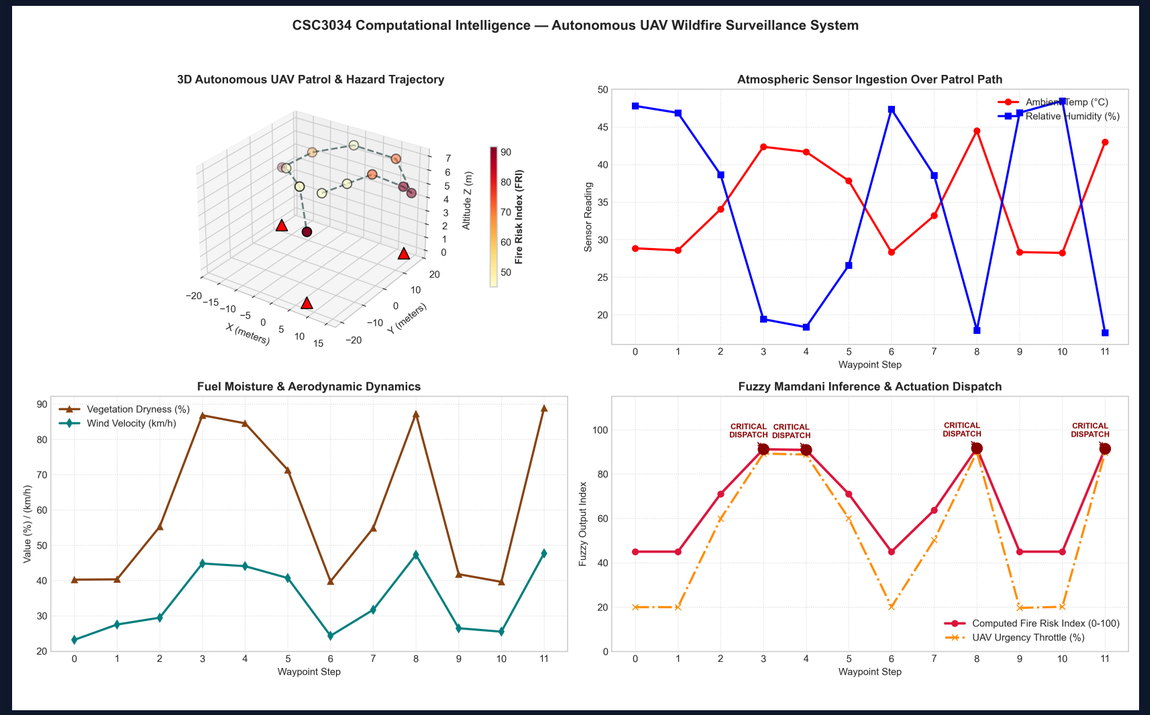
High-Fidelity Stage Highlights

- **Procedural Pine Forest:** 200m × 200m terrain populated with 48 pine trees with varied canopies and scales.
- **Thermal Hotspots:**
  - *Hotspot 01:* 38°C Smoldering brush fire
  - *Hotspot 02:* 78°C Active flaming core plume
  - *Hotspot 03:* 54°C Flare-up risk zone
- **Dual Camera Payloads:**
  - `ThermalGimbalCamera`: Downward nadir inspection
  - `ChaseCamera`: Dynamic 3rd-person follow cam for pilot view
- **RTX Ray Tracing Real-Time 2.0:** Sustained 72 FPS with stable GPU memory footprint on RTX 4060 Laptop GPU.

SIMULATION EXPERIMENTS

Autonomous Mission Telemetry & Actuation

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Observed System Behavior

- **Baseline Transit (Frames 0-100):** Ambient temp 24°C, FRI = 12.4% (SAFE). Drone sprints at 22.5 m/s to cover search area.
- **Hotspot 02 Incursion (Frames 200-340):** Sensor detects sudden thermal spike to 78.4°C. Fuzzy controller escalates FRI to **92.6% (CRITICAL)**.
- **Speed Throttling:** Urgency speed automatically drops to **6.5 m/s**, allowing the UAV to loiter and capture high-resolution imagery.
- **Hotspot 03 Flare-Up (Frames 450-520):** Moderate risk (FRI = 64.2%), drone decelerates to 12.0 m/s for confirmation.

COMPUTATIONAL INTELLIGENCE ANALYSIS

Defuzzification Comparative Benchmark

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Strategy Comparison

Method	Avg Latency	Actuator Smoothness	Noise Robustness
Centroid (COG)	0.884 ms	Exceptional (Smooth)	High
Bisector	0.912 ms	Good	Moderate
Mean of Max (MOM)	0.420 ms	Poor (Step Discontinuities)	Low

Finding: Although MOM computes faster, it produces sudden step-changes when peak membership shifts, which would cause violent motor torque spikes on a real quadrotor. Centroid was selected for flight safety.

Real-Time Edge Feasibility

- **Edge Platform:** Target deployment on NVIDIA Jetson Orin Nano (8 GB, 7-15 W power budget).
- **Timing Budget:** Fuzzy inference execution (0.884 ms) consumes only 1.7% of the 50 ms UAV flight control deadline (20 Hz loop).
- **Scalability:** Capable of evaluating 1,000+ rule inferences per second without throttling CPU flight controllers.

THANK YOU!

Project Defense & Demonstration Q&A Session

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GitHub: https://github.com/123haiku/csc3034-fuzzy-wildfire-sim